

Color spaces

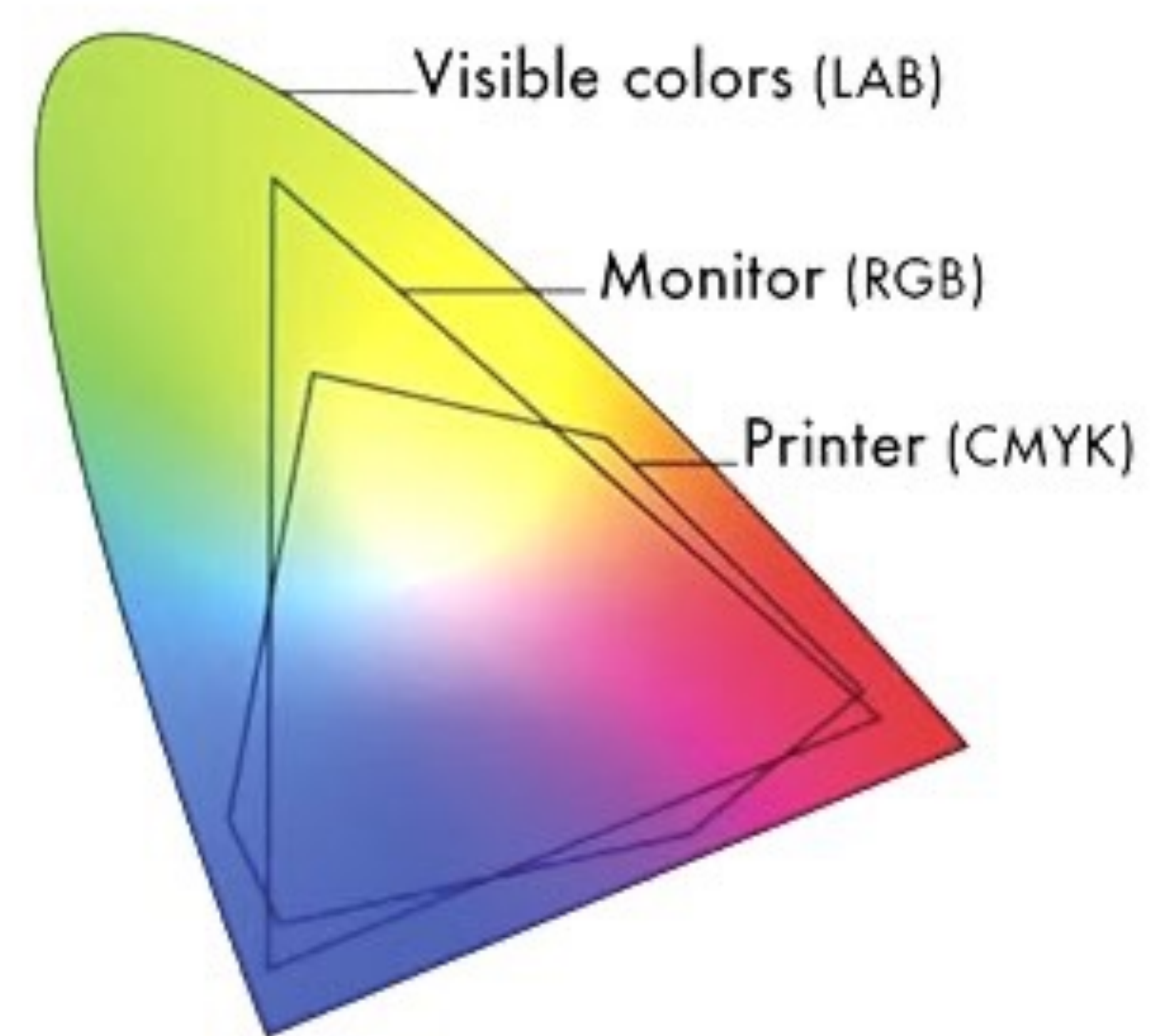
CIE $L^*a^*b^*$: perceptually uniform, changes in the values correspond more closely to perceived differences in color. Scaling of axes to represent "color distance"

RGB: digital displays and imaging

CMYK: printers; **HSL**, **HSV**, **HSK**, etc...

conversion between color spaces:

<https://medium.com/@alakhsharmacs/rgb-to-lab-color-space-conversion-formulas-insights-and-applications-7fd6efa519dd>



CIE $L^*a^*b^*$ space

L^* indicates lightness and a^* and b^* are chromaticity coordinates.

$+a^*$ is the red axis

$-a^*$ is the green axis

$+b^*$ is the yellow axis

$-b^*$ is the blue axis

Perceptually uniform: a nonlinear transformation of XYZ

